NullahNet — presenter card · Theme 3 · 10 min (aim 9:15, keep 45 s spare)

Before you start: present `zonal-deck.pdf` from your own laptop. Open the dashboard in a second tab with `bash deliverables/serve_dashboard.sh`, then go to http://localhost:8765 (it works offline except the map tiles). Backup slides 11–15: dashboard · how the twin works · limits · more Mumbai context · sources.

#	Time	Slide	Line to land
1	0:00–0:25	Title	"Not another barrier: a way to make the city's existing barriers accountable."
2	0:25–1:25	The leak, our plastic	"We target the plastic nobody is paid to collect."
3	1:25–2:30	One picture	"One operator decision changes: when the crew goes, and what proves what they removed."
4	2:30–3:25	Who intercepts	"We build on the city's booms and their verification norms."
5	3:25–4:20	Why booms leak (model)	"The twin tells us where to look; the tow tank must tell us whether it's true."
6	4:20–5:30	Evidence 1: timing	"In our model, a sensor without good timing can make things worse."
7	5:30–6:25	Evidence 2: retrofit what-ifs	"We only claim what the tow tank shows."
8	6:25–7:25	Evidence 3: sensing, records	"We report our weak numbers as plainly as our strong ones."
9	7:25–8:20	Conditions + cost	"Small add-on, large existing asset."
10	8:20–9:15	Deliver, pilot, pay	"Give us one existing boom to measure."

Numbers to remember

- 5 million kg a year: The Ocean Cleanup's survey *estimate* for the region.
- BMC booms in a 2025 report: 9 existing, expanding to 23, about ₹1 crore each. Maintenance contract: ₹14.5 crore for two years across six waterways.
- Model (SIMULATED): 92.5% of escaped plastic passes under or through the boom, but 51% of escape is an *assumed* still-water leak. Assumed curves, not findings.
- Full policy vs fill-fraction rule: −5.6%, lower in 39 of 40 *parameter settings*. Vs an assumed roster: −2.1%, small and uncertain.
- Fail-open alone: −0.1% (it's for safety). Pocket halving re-float: −5.0%; 30° boom: −15.0% (SENSITIVITY). Tide timing alone: −3.2%, doesn't clear zero.
- Sensor node: ₹83,481.98 in listed parts with enclosure. Installation and servicing are extra.

Fast answers

- *The Ocean Cleanup is here?* It covers two creek mouths, 61 to 92 tonnes a year expected. We propose to make BMC's booms measurable.
- *Simulated?* Yes, and it only chooses our first experiments. Real rain, archived forecasts and fitted tide; plastic behaviour is assumed.
- *Flooding?* The release is designed to open at a set head. Opening head, delay and reset are what test T3 measures. We don't claim it can't fail.
- *Novel trap?* No; floating tidal traps exist (Bandalong). What we propose is a retrofit of BMC's booms plus timing and records.
- *Camera at night?* Then fill is "unknown" and crews follow the roster. The ultrasonic measures water level, not fill.
- *Who pays?* The existing boom maintenance budget. EPR is upside: drain-plastic eligibility is not settled.

Don't say: "tamper-proof" · "blockchain" · "can't flood" · "seasons" (say "parameter settings") · "the booms have no sensors" · absolute tonnes from the twin · "our trap is new" · any guilt in the Mithi case · "we built the hardware" (it's PLANNED).